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Wildfire detection using DenseNet model.

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MSc FPR Declaration

This report is submitted in partial fulfilment of the requirement for the degree of:

Master of Science in 7COM1039-0206-2024 - Advanced Computer Science Masters Project, at the University of Hertfordshire (UH).

I hereby declare that the work presented in this project and report is entirely my own, except where explicitly stated otherwise. All sources of information and ideas, whether quoted directly or paraphrased, have been properly referenced in accordance with academic standards. I understand that any failure to properly acknowledge the work of others could constitute plagiarism and may result in academic penalties.

I did not use human participants in my MSc Project.

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Abstract

Wildfires represent a significant danger to ecosystems, economy, and human safety, with their occurrence increasing owing to climate change and anthropogenic activities. Prompt and precise detection is crucial to mitigate damage and facilitate efficient emergency actions. This study presents a deep learning-based wildfire detection system that incorporates sophisticated image preprocessing techniques, including Multi-Scale Retinex (MSR) and RGB-to-YCbCr color space translation, with the DenseNet121 architecture. The Wildfire Dataset, consisting of various fire and non-fire photos, was utilized for model training and assessment. Exploratory Data Analysis (EDA) identified significant difficulties including brightness fluctuation, class imbalance, and resolution discrepancies. Results indicate that DenseNet with preprocessing attained an impressive 98.84% test accuracy, in contrast to 90.95% without preprocessing, reflecting an 8% increase. The accuracy of the CNN model increased from 82.89% to 92.11% following preprocessing. Preprocessing significantly improved contrast, feature visibility, and resilience, resulting in substantial decreases in misclassification rates. The results confirm that improving input quality by MSR and color space conversion directly enhances classification performance, particularly under difficult visual situations like smoke, haze, and inadequate illumination. This study enhances image-based wildfire detection capabilities, providing a more precise and dependable solution than conventional raw-image classification methods.

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Section 1: Introduction

Forests are essential for maintaining ecological equilibrium, sometimes referred to as the "lungs of the planet" due to their capacity to absorb CO_2 and emit O_2. They promote biodiversity, manage hydrological cycles, protect crops from wind, and enhance national economies through resources and job opportunities. These ecosystems are increasingly imperilled by wildfires—unrestrained natural calamities that have escalated in frequency and intensity due to climate change and anthropogenic activity. (Barmpoutis, P., et al,2020)

Wildfires annually devastate millions of acres, adversely affecting wildlife, agricultural output, and human lives. Prominent instances encompass the 2020 Australian bushfires and the 2018 California wildfires. Although climate change exacerbates wildfire conditions, research indicates that around 85% of U.S. wildfires from 1992 to 2015 were attributable to human activity. (Bouguettaya, A., et al,2022)

The essential factor in mitigating wildfire damage is prompt discovery. Nevertheless, conventional technologies such as satellites and watchtowers are constrained by constraints in visibility, expense, and resolution (Barmpoutis, P., et al,2020). This necessitates the development of novel and dependable technology for the early and precise detection of wildfires, facilitating prompt responses and reducing environmental and economic damage.

1.1 Contemporary Challenges

Notwithstanding continuous developments, wildfire detection systems encounter several significant hurdles. Satellite-based methods, despite their extensive coverage, can have inadequate spatial resolution and prolonged return intervals, hindering the detection of minor or nascent fires particularly in the presence of cloud cover or deep forest canopies (Ororatech ,2023). Ground-based techniques such as watchtowers and infrared cameras are constrained by visibility range, substantial installation expenses, and environmental disruptions, especially in adverse weather conditions or at night.

Moreover, several systems depend on human oversight, which is susceptible to delays and inaccuracies resulting from exhaustion or insufficient real-time response. Detection systems frequently encounter false positives, mistakenly identifying sunlight reflections or industrial heat sources as genuine fires, so undermining faith and taxing emergency resources. (Johnston, J.M., et al, 2018)

These concerns underscore the necessity for sophisticated, scalable, and accurate wildfire detection techniques. This study mitigates these deficiencies by using picture preprocessing and DenseNet-based deep learning to improve classification accuracy and reduce detection latency.

1.2 Aim of the project

This research aims to create a reliable and precise wildfire detection system that incorporates advanced picture preprocessing methods specifically Multi-Scale Retinex (MSR) and RGB-to-YCbCr color space transformation alongside the DenseNet deep learning architecture for classification. This method aims to increase the quality of wildfire images, boost feature

extraction, and attain high-precision classification under various environmental circumstances, including smoke, haze, and fluctuating light. The primary objective is to facilitate prompt and accurate wildfire detection, minimize false alarms, and enhance timely response initiatives.

Novelty of the project

This study examines a significant deficiency in wildfire detection studies. Although several deep learning models, have been utilized for this issue, most methods depend on unrefined or lightly processed images. These models frequently have difficulties in detection under conditions of smoke, haze, or inadequate lighting, resulting in false positives and delayed responses.

This study is innovative because to its hybrid methodology, integrating advanced image preprocessing techniques (MSR and color space transformation) with the DenseNet model to improve feature visibility and extraction. This integration seeks to enhance detection precision while minimizing false alarms. The study provides value by conducting a comparison analysis of DenseNet with alternative topologies, yielding insights into scalability, accuracy, and practical use in wildfire detection systems.

1.3 Research questions

- What is the effect of using Multi-Scale Retinex (MSR), color space transformation (RGB to YCbCr), and DenseNet on the accuracy and robustness of wildfire detection relative to prior studies?
- What is the extent to which photo preprocessing techniques such as Multi-Scale Retinex (MSR) and colour space conversion can enhance visibility, reduce noise, and improve the classification accuracy of wildfire detection models?
- How does the DenseNet model improve feature extraction and classification accuracy in wildfire detection compared to other deep learning architectures?

1.4 Objectives of the project

- To do a comprehensive literature analysis of current wildfire detection techniques, highlighting significant shortcomings in conventional systems and contemporary deep learning models, with an emphasis on preprocessing deficiencies and classification obstacles.
- To create a wildfire detection system utilizing the DenseNet architecture, incorporating sophisticated picture preprocessing methods to improve feature extraction and classification precision.
- To implement and evaluate Multi-Scale Retinex (MSR) and RGB-to-YCbCr color space transformation for enhancing picture clarity, contrast, and segmentation of fire regions in adverse environmental circumstances.
- To assess the proposed model's efficacy through precision, recall, F1-score, and accuracy, while reducing false positives under several wildfire situations.
- To evaluate DenseNet in comparison to various deep learning architectures from literature and to examine the system's scalability, robustness, and practical usefulness.

1.5 Feasibility, Commercial Context, and Risk

Feasibility and scope

This project aims to create a wildfire detection system utilizing image-based deep learning methodologies, particularly the DenseNet architecture. The system employs sophisticated preprocessing techniques—Multi-Scale Retinex (MSR) and RGB-to-YCbCr color space conversion—to augment picture quality and refine fire segmentation. The project is achievable within the academic schedule and resources by utilizing publicly accessible wildfire datasets.

The scope encompasses model training, testing, assessment, and comparison with other deep learning models. The proposal excludes real-time detection, IoT integration, drone or satellite deployment, and fire management operations. This study is exclusively focused on assessing the efficacy of preprocessing-augmented DenseNet models for the categorization of wildfire images.

Commercial and Economic Environment

The market for wildfire detection systems is seeing significant expansion. The global market for forest wildfire detection systems was valued at around USD 712.2 million in 2023 and is anticipated to reach USD 1.20 billion by 2031, with a compound yearly growth rate (CAGR) of 5.83% over the forecast period. This expansion is propelled by the rising frequency of wildfire occurrences and the resulting need for sophisticated detection methods. (Joint Economic Committee, 2023)

In the United States, climate-induced wildfires are projected to incur yearly costs ranging from $394 billion to $893 billion, representing between 2–4% of the nation's GDP. In Europe, wildfires have similarly caused substantial financial harm. In 2023, wildfires caused €4.1 billion (£3.46 billion) in damages, predominantly affecting nations like as Greece, Spain, and Italy. These expenses include property loss, health effects from smoke, and extensive economic disruptions. (The Week, 2023)

This research aims to improve image-based wildfire detection using sophisticated preprocessing and deep learning, in accordance with industry demands.

Risk Analysis

The risk register identifies six key risks (table 1):

Table 1: Risk assessment of the project.

Risk ID	Risk Description	Impact	Likelihood	Mitigation Strategy
R1	Dataset Limitations – Insufficient wildfire images, unbalanced dataset, or lack of diverse environmental conditions	High	Medium	Use data augmentation techniques (geometric transformations, brightness adjustments) and incorporate publicly available datasets
R2	Poor Image Preprocessing Effectiveness – MSR or colour space conversion may not enhance visibility as expected	Medium	Medium	Perform fine-tune parameters for best performance
R3	Overfitting of Model – Small training dataset may lead to poor generalization to real-world scenarios	High	Medium	Use cross-validation, dropout regularization, and extensive data augmentation to improve generalization
R4	Hyperparameter Optimization Complexity – Finding the best parameters for DenseNet may be time-consuming	Medium	Medium	Use automated hyperparameter tuning tools like GridSearchCV or Bayesian Optimization to optimize model performance efficiently
R5	Hardware Constraints – Computational requirements for training DenseNet may be high	High	Low	Use Google Colab Pro, AWS, or GPU-enabled environments for efficient model training
R6	Final VIVA Preparation Risk – Time constraints in finalizing documentation, report, and presentation	Medium	Medium	Maintain weekly progress tracking and allocate sufficient time for report writing and VIVA preparation

1.6 Report Structure

This report is structured into six principal sections.

- Section 1 presents the project's origins, aims, innovation, and viability.
- Section 2 presents an extensive literature study, highlighting deficiencies in existing wildfire detection techniques and substantiating the suggested methodology.
- Section 3 delineates the approach, encompassing dataset specifications, preprocessing methods, model construction, and assessment framework.
- Section 4 delineates the experimental outcomes, encompassing performance comparisons across models and the effects of preprocessing.
- Section 5 delineates principal findings, aligns results with study objectives, acknowledges limits.
- Section 6: Suggests avenues for future research, ending in a conclusion that underscores the project's relevance and contributions.

Section 2: Literature review

Wildfire detection is a vital study area propelled by the rising frequency and intensity of wildfires worldwide. Conventional techniques, such as satellite images, heat sensors, and terrestrial watchtowers, frequently fail to provide prompt, high-resolution, and economical detection especially in environments influenced by smoke, haze, or variable illumination. Consequently, deep learning and computer vision methodologies have surfaced as viable options for automating wildfire identification using picture data.

Convolutional Neural Networks (CNNs) like ResNet, VGG, and MobileNet have been extensively utilized in the categorization of wildfire images. Nevertheless, several models depend on unrefined or weakly refined pictures, hence constraining their robustness and precision in practical applications. This approach mitigates that constraint by integrating sophisticated preprocessing techniques Multi-Scale Retinex (MSR) for light correction and RGB-to-YCbCr color space transformation for improved fire feature isolation into a DenseNet-based classification framework.

2.1 Related works

Wu, et al. (2020) introduced a wildfire detection system employing MobileNetV2 and transfer learning for rapid and precise picture categorization. The model was trained using wildfire photos acquired by UAVs and included a bespoke data augmentation pipeline that simulated difficult circumstances such as Gaussian blur and noise. The UAVs provide versatility, security, and economic efficiency relative to conventional detecting methods. The collection comprised annotated aerial pictures featuring both fire and non-fire occurrences. The model attained 96.7% accuracy on the original dataset, 95.3% on blurred photos, and 95.3% on images with Gaussian noise. This exhibits significant generality and resilience across many visual contexts, highlighting its applicability for real-time wildfire monitoring systems in practical settings.

Treneska, and Stojkoska, (2021) employed transfer learning with ResNet50 to identify wildfires from photos captured by UAVs. Their methodology optimized many pre-trained CNNs, such as ResNet50, Xception, and InceptionV3, utilizing the FLAME dataset, which comprises approximately 39,000 annotated drone-captured photos of fire and non-fire scenarios. Fine-tuning ResNet50 achieved the highest classification performance with an accuracy of 88%, representing an 11% enhancement over baseline models. The researchers established that transfer learning markedly improves generalization and is appropriate for real-time wildfire detection. Their research underscored the increasing significance of UAV-based data acquisition, especially for high-resolution, preliminary fire monitoring applications.

Ghali, et al. (2022) introduced an innovative ensemble wildfire detection system that integrates EfficientNet-B5, and vision transformers—TransUNet and TransFire—for classification and segmentation purposes. Their dataset comprised UAV-acquired aerial photos, annotated for fire and non-fire areas. The classification model attained an accuracy of 95.12%, These models tackled issues such as limited fire size, visual deterioration, and intricate backdrops. This work

confirms the efficacy of integrating CNNs and transformers for precise fire localization and detection in UAV surveillance applications.

Sathishkumar, et al. (2023) employed transfer learning with VGG16, InceptionV3, and Xception, augmented by Learning without Forgetting (LwF) to preserve model efficacy on previous datasets. The research utilized the BowFire dataset for novel tasks and a distinct foundational dataset. The Xception model achieved 79.23% accuracy on BowFire without LwF. Utilizing LwF, the performance enhanced to 91.41% while preserving 96.89% on the original dataset. This methodology mitigates catastrophic forgetting in transfer learning and is effective for adapting wildfire models to novel geographies or data distributions. It is especially advantageous when new training data is scarce, while preserving historical performance is essential.

The CT-Fire model, developed by Ghali, and Akhloufi, (2023), is a hybrid technique that integrates CNN and Transformer architectures, specifically utilizing RegNetY and EfficientFormer v2. The training and testing utilized a combination of terrestrial and airborne wildfire datasets: FLAME, CorsicanFire, DeepFire, and Fire. CT-Fire attained 97.62% accuracy on terrestrial pictures and 87.77% on aerial photos, surpassing benchmark CNNs and transformers. The ensemble exhibited resistance to intricate backdrops, minor burning areas, and fluctuating fire intensities. This study demonstrates that the integration of CNNs and transformers effectively captures both local and global picture data, rendering it appropriate for thorough wildfire detection across diverse imaging situations.

Wang, et al, (2023) developed two wildfire detection modules: a Reduce-VGGNet for classification and an improved CNN that integrates spatial and temporal information for regional detection. The research sought to reconcile performance with accessibility in educational settings. The project utilized a selected wildfire dataset appropriate for Machine Vision course applications. The Reduce-VGGNet attained a classification accuracy of 91.20%, but the spatio-temporal CNN acquired an accuracy of 97.35% in region detection. This modular methodology delivers both scholarly and practical benefits, presenting a dependable detection system characterized by high precision, while being appropriate for educational purposes.

Jonnalagadda, and Hashim, (2024) presented SegNet, a segmented convolutional neural network designed for real-time wildfire identification utilizing drone (UAV) footage. The approach emphasizes enhancing processing performance by minimizing superfluous feature maps while preserving pixel-level accuracy for amorphous entities such as fire and smoke. A bespoke UAV wildfire dataset was utilized, obtained through drone flights over fire-prone regions. SegNet markedly enhanced inference speed and attained exceptionally high detection accuracy; although the precise numerical precision was not specified, it outperformed conventional CNN methods in both performance and computing economy. This research offers a significant addition to real-time wildfire monitoring, particularly in bandwidth-limited or mobile environments.

Yandouzi et al. utilized deep transfer learning with several pre-trained CNN architectures, including VGG16, ResNet50, Xception, MobileNet, and NASNetMobile, for the detection of forest fires. The scientists trained their models using a composite wildfire dataset obtained from

UAVs, satellites, and terrestrial platforms. Several models attained detection accuracy over 97.1%, demonstrating exceptional generalization. Their comparative analysis highlighted DenseNet and InceptionResNetV2 as particularly formidable candidates. The study emphasized the practical use of transfer learning, demonstrating that excellent accuracies were attained even with very limited training datasets. This renders it appropriate for practical applications using constrained labelled wildfire data.

2.2 Critical Assessment of existing Literature

The analysed papers together demonstrate substantial advancements in wildfire detection via the implementation of deep learning, specifically utilizing transfer learning and CNN-based frameworks. Numerous studies, including those by Wu et al. (2020), Treneska and Stojkoska (2021), and Ghali et al. (2022), illustrate that pretrained models such as MobileNetV2, ResNet50, and EfficientNetB5 are very proficient when fine-tuned for fire classification tasks. The use of vision transformers, shown by CT-Fire and TransFire, improves spatial awareness and feature representation, facilitating precise detection in congested or intricate visual environments. Numerous models, such those introduced by Wang et al. (2023) and Jonnalagadda and Hashim (2024), emphasize computing efficiency, rendering them appropriate for real-time UAV deployment. Moreover, techniques like as Learning without Forgetting (LwF) demonstrate potential in preserving knowledge during cross-domain adaptation, and research by Yandouzi et al. (2022) validates that deep models may maintain remarkable performance while having minimal labeled data.

Notwithstanding these developments, a significant drawback in most investigations is the inadequate application of sophisticated picture preparation techniques. Although several articles utilize fundamental data augmentation techniques (e.g., rotations, flips, noise), few include more sophisticated approaches like as denoising filters, contrast improvement, Retinex algorithms, or color space transformations. This absence may diminish model robustness in real-world scenarios marked by smoke, haze, low illumination, or inadequate contrast. Numerous approaches depend significantly on architectural complexity to get performance improvements, which, although beneficial in controlled datasets, may not translate effectively to noisy or damaged field data. Research conducted by Wu et al. (2020) and Ghali et al. (2022) demonstrates commendable performance on pristine datasets; nevertheless, there is little exploration into the potential enhancements in accuracy and resilience through preprocessing techniques. The prevailing tendency indicates a disparity in the synergy between the increase of input quality and classification, highlighting an opportunity for additional investigation such as this project which seeks to rectify this constraint by incorporating certain preprocessing techniques with deep learning models.

2.3 Gaps Identified in the Literature

Current work on wildfire detection via deep learning has achieved significant advancements; nonetheless, notable gaps persist, especially concerning picture preprocessing and the equilibrium between model complexity and efficiency. This study tackles these deficiencies by employing an innovative pipeline that combines sophisticated preprocessing with an efficient, lightweight model.

Insufficient Advanced Image Preprocessing Techniques: Numerous research depends on fundamental data augmentation methods such as rotation, flipping, and Gaussian noise (Govil, et al 2020). Nonetheless, these methodologies fail to sufficiently equip models for practical situations characterized by haze, smoke, or inadequate lighting. This effort mitigates this deficiency by employing Multi-Scale Retinex (MSR) for lighting correction and transforming RGB to YCbCr color space, therefore markedly improving picture clarity and contrast, which renders the fire zones more discernible prior to classification.

Excessive Dependence on Deep Architectures Without Input Augmentation: Architectural advancements like TransUNet or CT-Fire prioritize model depth at the expense of input quality. This research integrates a somewhat intricate yet effective architecture (DenseNet) with specific preprocessing, demonstrating that input augmentation can increase accuracy without necessitating computationally intensive model layers. (Jain, et al 2020)

Insufficient Model Efficiency in Real-Time Contexts: Numerous high-performing models are computationally demanding and inappropriate for real-time or embedded implementation. (Saleh, et al,2024)

This project utilizes DenseNet's efficiency, since its dense connection minimizes duplicate calculations and enhances feature reuse. This guarantees rapid, memory-efficient inference, rendering it optimal for real-time UAV-based wildfire surveillance systems.

2.4 Relation to Research and Hypothesis

The examined literature demonstrates substantial advancements in wildfire identification utilizing deep learning and images obtained from UAVs. Research conducted by Wu et al. (2020), Treneska and Stojkoska (2021), and Ghali et al. (2022) has effectively utilized transfer learning methodologies on CNN architectures, including MobileNetV2, ResNet50, and EfficientNet-B5, to achieve significant accuracy in wildfire detection. Other models, such as CT-Fire and SegNet, have prioritized architectural innovation and computing efficiency to satisfy real-time operating requirements.

Nonetheless, a significant weakness in these investigations is the inadequate emphasis on improving the visual quality of input data by sophisticated preprocessing approaches. Many models predominantly depend on data augmentation or increased architectural depth to enhance performance, neglecting the advantages of techniques like as denoising, illumination correction, or color space transformation in intricate, real-world settings characterized by smoke, haze, and inadequate lighting.

This study directly tackles this deficiency by including picture preprocessing techniques—specifically Multi-Scale Retinex (MSR) and RGB-to-YCbCr color space transformation—into the classification pipeline. The modifications are designed to enhance picture clarity and feature differentiation prior to processing by the DenseNet model, recognized for its effective feature propagation and reutilization.

This effort enhances previous breakthroughs in deep learning for wildfire detection while introducing a more equitable and resilient methodology that enhances classification performance under compromised visual circumstances. It aligns with the overarching aims and objectives of the study, which encompass minimizing false positives, enhancing detection precision, and guaranteeing the model's efficiency and appropriateness for real-time deployment contexts, such as UAV-based surveillance.

Section 3: Methodology of the project

This section delineates the methodical strategy employed to create an effective and resilient wildfire detection system utilizing deep learning, along with sophisticated image preprocessing techniques. The methodology aims to improve image clarity and classification precision in real-world visual problems, including smoke, haze, and variable lighting—conditions frequently inadequately represented in current models.

The workflow, depicted in Figure 1, commences with data collection, succeeded by exploratory data analysis (EDA) to comprehend data distribution, detect class imbalances, and visualize sample attributes. Thereafter, the raw pictures undergo preprocessing through RGB-to-YCbCr color space translation to improve the distinction between fire and non-fire features, succeeded by Multi-Scale Retinex (MSR) processing for illumination normalization and contrast augmentation.

Upon completion of preprocessing, the augmented photos are utilized for model construction, wherein the DenseNet architecture is trained to classify the existence of wildfires. The model is subjected to hyperparameter adjustment to enhance performance. The concluding phase entails performance assessment utilizing metrics such as accuracy, precision, recall, and F1-score to evaluate model efficacy under diverse environmental settings.

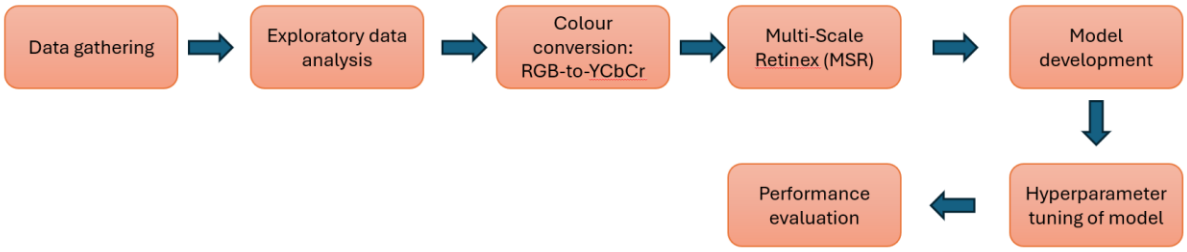


Figure 1: Workflow of the project.

3.1 Dataset Gathering

This project utilizes the Wildfire Dataset compiled by El-Madafri et al. (2023), consisting of 2700 high-resolution RGB photos depicting diverse wildfire and non-wildfire scenarios. The images derive from various public sources, including government repositories, Flickr, and Unsplashed, encompassing a wide range of environmental contexts—such as different forest types, geographic regions, fire intensities, and complicating visual elements (e.g., clouds, sun glare, or fog). The dataset is organized into two main categories fire and nofire.

Link to download dataset: **https://www.kaggle.com/datasets/elmadafri/the-wildfire-dataset**

Publication link: **https://www.mdpi.com/1999-4907/14/9/1697**

The dataset was developed to overcome significant shortcomings in current forest fire detection datasets (figure 2): restricted diversity, inadequate representativeness, and poor management of visual confounders. These problems frequently result in limited generalizability and

elevated false positive rates in deep learning models. This resource enhances training realism for AI detection systems by providing a dataset with diverse topography, weather conditions, camera angles, and aspects of fire and smoke. (El-Madafri et al. 2023)

Figure 2: Sample images from the dataset.

Ethical considerations were meticulously addressed: all photos were obtained from the public domain or open-licensed platforms, assuring adherence to copyright and data-sharing legislation. This legal and ethical transparency facilitates repeatability and promotes open scientific inquiry. All pictures in the dataset are either open-source or publicly licensed, devoid of personally identifying information (PII), and are intended exclusively for academic and research purposes. The dataset is published under the Creative Commons Attribution 4.0 International License (CC BY 4.0), allowing unrestricted use, distribution, and modification with proper attribution to the authors. Additional details regarding the license are available at https://creativecommons.org/licenses/by/4.0/.

This dataset was chosen for the study because of its significant diversity, organized annotations, and its congruence with the project's aim: to enhance wildfire detection efficacy by image preprocessing and robust classification. It functions as a standard for practical applicability and facilitates testing under demanding visual situations, rendering it optimal for assessing preprocessing-augmented deep learning models. (El-Madafri et al. 2023)

3.2 Exploratory Data Analysis

Class distribution of the dataset

An exploratory data analysis was performed to comprehend the dataset's structure and detect potential biases, concentrating on class distribution among the training, validation, and test sets. The pie charts above illustrate the distribution of photos categorized as fire and nofire within each subset.

The dataset exhibits a moderate class imbalance, with 61.3% of photos categorised as nofire and 38.7% as fire. This disparity is consistently seen across all subsets (figure 3):

- Training Set: 61.3% non-fire, 38.7% fire
- Validation Set: 61.0% non-fire, 39.0% fire
- Test Set: 61.4% non-fire, 38.6% fire

This division of data partitions guarantees equitable model training and assessment, reducing the likelihood of overfitting or biased performance favouring the majority class (Vasconcelos, et al,2024). Nonetheless, the minor class imbalance may still affect the model's propensity to predict the nofire class more frequently, rendering precision and recall essential evaluation metrics.

This imbalance will be addressed during training using strategies such image augmentation, to provide strong learning and precise detection in both fire and no-fire scenarios.

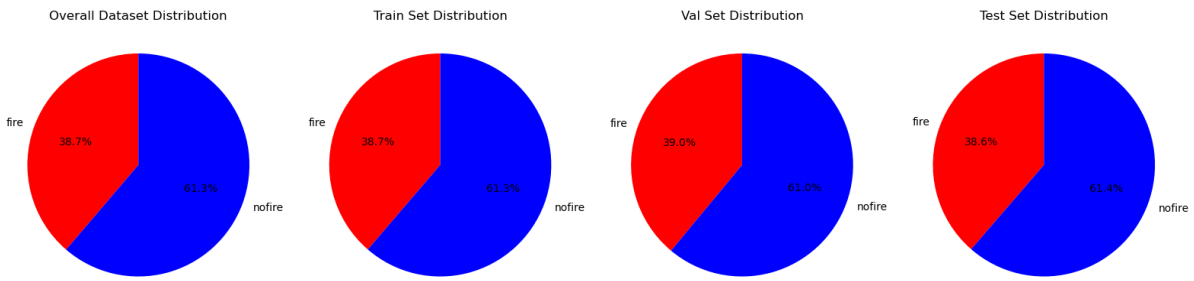


Figure 3: Class distribution of the dataset.

Image size distribution

The examination of image size distribution (figure 4) indicates a broad spectrum of resolutions in both the fire and nofire categories, with dimensions ranging from under 500×500 pixels to exceeding 10,000 pixels in both width and height. Images of fires typically aggregate at regular resolutions, presumably originating from consistent sources like UAVs or surveillance cameras. Conversely, nofire photos demonstrate enhanced dimensional diversity, signifying a range of image sources and content types.

This research was crucial for guiding preprocessing options, as convolutional neural networks such as DenseNet necessitate fixed input dimensions. It facilitates the selection of an optimal resizing dimension that preserves critical visual attributes while preserving computational efficiency. It also underscores potential biases particular to classes in resolution that may impact model generalization.

Standardizing image dimensions via scaling or padding guarantees uniform model input, reduces distortion, and prevents GPU memory overload. This stage is essential for attaining balanced training efficacy and dependable wildfire detection under diverse environmental conditions.

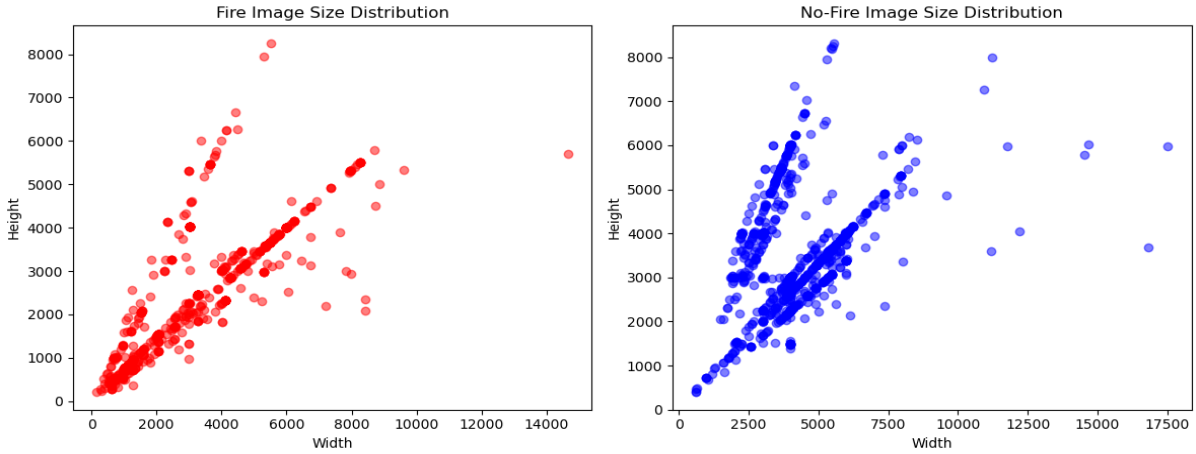


Figure 4: Image size distribution of the dataset.

Colour Distribution Analysis

To further examine the visual attributes of the dataset, average RGB histograms were generated for both the fire and no-fire image categories. This research identifies color intensity patterns distinct to each class, which can drive preprocessing choices and facilitate the development of feature-sensitive classification models.

The histogram for fire photos exhibits a reasonably similar distribution across the three channels (red, green, and blue), with a little predominance in the red channel at elevated pixel intensities. This is anticipated because to the existence of flames that emit high-energy red and orange light. Furthermore, fire scenes frequently feature smoke and intense illumination, which can elevate the intensity in the higher pixel value ranges (approaching 250).

Conversely, the nofire photos demonstrate a more pronounced prevalence of lower pixel brightness, particularly within the blue and green channels. This suggests that these pictures may feature increased flora, sky, and shaded areas, leading to a cooler and darker overall tone in contrast to fire scenes. A notable peak exists at both the low and high extremes of the spectrum, likely attributable to shadowed regions and overexposed backgrounds, respectively.

The histogram patterns (figure 5) confirm the necessity for illumination correction and color normalization, particularly for fire photographs that may experience glare, as well as for non-fire situations affected by shadows or underexposure. This endorses the application of Multi-Scale Retinex (MSR) and RGB-to-YCbCr conversion in the methodology. Comprehending channel behavior aids in recognizing distinguishing characteristics. The predominance of red in fire imagery can be used via colour space transformation to enhance pertinent fire-related characteristics while diminishing background interference.

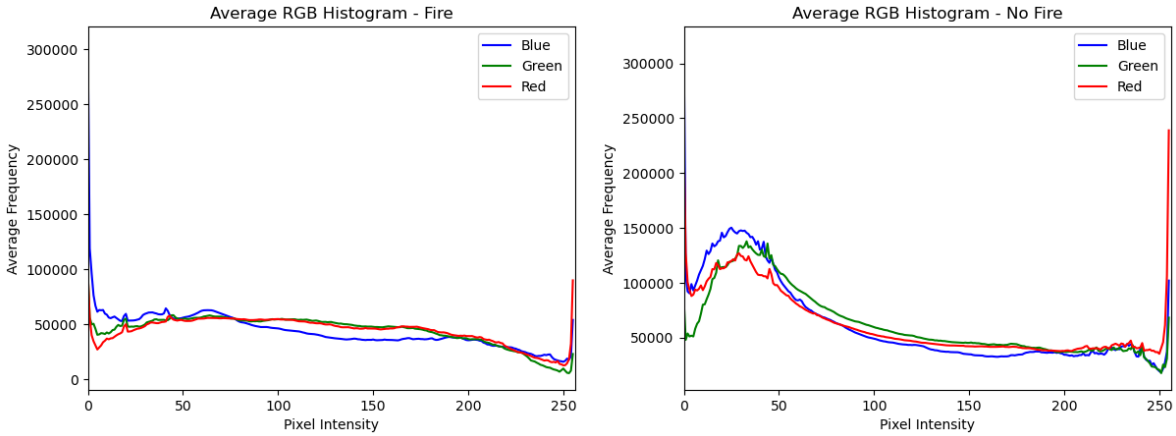


Figure 5: Colour Distribution Analysis of each class.

Brightness, Contrast, and Standard Deviation Analysis

To further examine the visual diversity within the dataset, three essential image statistics were analysed: brightness, contrast, and standard deviation (figure 6). These characteristics directly influence the appearance of fire and non-fire areas in images and are essential for assessing the efficacy of preprocessing techniques such as Multi-Scale Retinex (MSR).

Brightness Distribution

The histogram contrasting brightness levels of fire and non-fire photos indicates that fire photographs generally exhibit elevated brightness values, predominantly grouping between 110 and 160. This is anticipated, as fire produces light and frequently results in overexposed areas in photographs. Conversely, nofire photos display diminished brightness levels, predominantly between 50 and 110, reflecting natural lighting situations like shadows, foliage, or overcast skies.

Observation: The distinct brightness differentiation among classes facilitates the application of brightness-sensitive preprocessing techniques such as MSR, which may standardize illumination across varying conditions while maintaining significant visual indicators associated with fire.

Contrast Distribution

The contrast histogram reveals a moderate overlap between fire and no-fire images, with both distributions peaking in the 50–65 range. Conversely, nofire photos demonstrate a broader distribution, revealing greater contrast variance. This may result from the existence of many textures, including trees, sky, water, or ground, which frequently reflect light unevenly.

Observation: Although contrast alone may not serve as a robust differentiator, augmenting local contrast, as implemented in MSR, might enhance edge identification and emphasize fire-related textures, particularly in instances of small or faint fires.

Standard Deviation Distribution

The analysis of standard deviation, which quantifies the variability of pixel intensities, closely resembles the distribution of contrast. Once more, nofire photos exhibit a wider spread, signifying greater variety in their pixel values. Fire photos are typically more concentrated in the mid-range, perhaps because the dominance of flames diminishes variety.

Observation: Comprehending intensity variability aids in predicting the model's capacity to manage intra-class fluctuations. MSR and color space modifications can diminish this variability, enhancing feature consistency for model training.

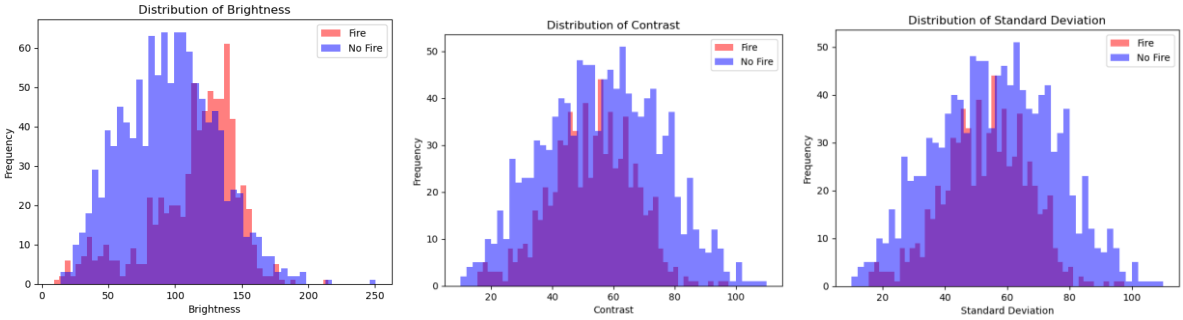


Figure 6: Brightness, Contrast, and Standard Deviation Analysis of the dataset.

These findings unequivocally endorse the project's decision to employ illumination correction (MSR) and color space conversion to augment feature visibility.

TSNE Analysis

To illustrate the separability of fire and no-fire photos based on extracted characteristics, t-distributed Stochastic Neighbor Embedding (t-SNE) was utilized on brightness, contrast, texture metrics, and mean RGB values (Devassy, et al,2020). The resultant 2D plot (figure 7) exhibits partial grouping of the two classes, accompanied by some overlap, suggesting that although the characteristics possess discriminative capability, they are inadequate for definitive class separation.

The overlap indicates that specific nofire photographs may exhibit fire-like characteristics (e.g., intense sunshine or autumn foliage), whereas certain fire images may appear visually understated. This underscores the necessity for sophisticated preprocessing methods—such as Multi-Scale Retinex and RGB-to-YCbCr conversion—to improve contrast and feature definition.

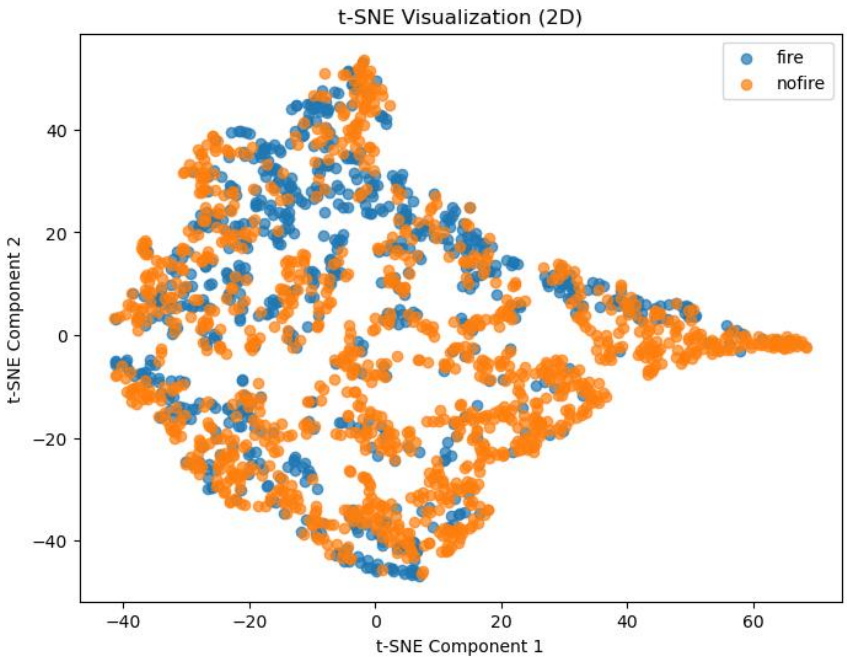


Figure 7: TSNE plot of the dataset.

The result further endorses the application of DenseNet, a deep learning model proficient in learning non-linear feature connections that t-SNE projections indicate are challenging to capture with simple statistical features alone. This phase confirms essential methodological choices in feature augmentation and model selection.

Conclusive Observations from Exploratory Data Analysis

The Exploratory Data Analysis (EDA) uncovers several essential findings that guide the project's design. The dataset exhibits a moderate class imbalance (61% no fire, 39% fire), requiring class balancing methods such as augmentation. Substantial discrepancies in image dimensions within classes underscore the necessity for resizing to maintain uniformity and enhance model efficiency. Analysis of the RGB histogram indicates a predominance of the red channel in fire photos, so endorsing the application of color space modification (e.g., RGB-to-YCbCr) to improve the visibility of fire features. Analyses of brightness and contrast reveal that fire photos are often brighter, but nofire images exhibit greater fluctuation, underscoring the necessity for lighting adjustment by Multi-Scale Retinex (MSR). The standard deviation indicates increased visual complexity in nofire scenarios. The t-SNE plot indicates partial clustering and class overlap, so affirming the appropriateness of DenseNet for discerning non-linear patterns that handcrafted features struggle to differentiate. These insights collectively validate the selected preprocessing methods and model architecture for effective wildfire identification.

3.3 Data preprocessing: RGB-to-YCbCr

The RGB-to-YCbCr conversion is a prevalent method in image processing that distinguishes luminance (brightness) from chrominance (colour) information (Safir, P., 2022). This transformation separates the original RGB image into three components:

- Luminance (brightness)
- Cb: Chrominance-blue (blue subtracted from luminance)
- Cr: Chrominance-red (red subtracted from luminance)

Conversion Formula

Equation 1: Formula for conversion of RGB-to-YCbCr

The standard RGB-to-YCbCr transformation (ITU-R BT.601 standard) is:

$$
\begin{aligned}
Y &= 0.299R + 0.587G + 0.114B \\
Cb &= 128 - 0.168736R - 0.331264G + 0.5B \\
Cr &= 128 + 0.5R - 0.418688G - 0.081312B
\end{aligned}
$$

By segregating luminance from chromatic elements, the model may concentrate on structural and brightness-related fire characteristics (such as flame contours and smoke) without being diverted by extraneous colour fluctuations in the background. (Safir, P., 2022)

The function convert_rgb_to_ycbcr(image) use created which uses OpenCV's inherent color space conversion capabilities to convert an image from the RGB color space to the YCbCr color space.

Rationale for Selection

Fire detection frequently necessitates heightened sensitivity to light intensity and form rather than just hue. In RGB, color and luminance are interdependent, which can provide challenges in environments with sunlight, foliage, or reflections.

The RGB-to-YCbCr conversion alleviates this problem by distinguishing luminance (Y) from chromatic components (Cb, Cr), hence improving the model's emphasis on flame intensity and structure. (Fang, et al,2025)

What Enhancements It Provides

The RGB-to-YCbCr conversion improves fire detection by separating the luminance component (Y), enabling the model to concentrate on structural fire patterns instead of being affected by background colour fluctuations. This minimizes false positives from red or orange items that visually mimic flames in RGB space.

Figure 8 demonstrates that the YCbCr-transformed fire image accentuates areas of high luminosity and form related to smoke and flames, while diminishing extraneous colour information. Likewise, in the non-fire scenario, the change mitigates distracting colours from flora and landscape, providing a more consistent backdrop. The clarity in luminance and diminished chromatic noise facilitate the classifier's ability to differentiate between fire and non-fire circumstances.

RGB-to-YCbCr enhances edge definition, flame intensity recognition, and background suppression, all essential for augmenting model accuracy across varied environmental contexts.

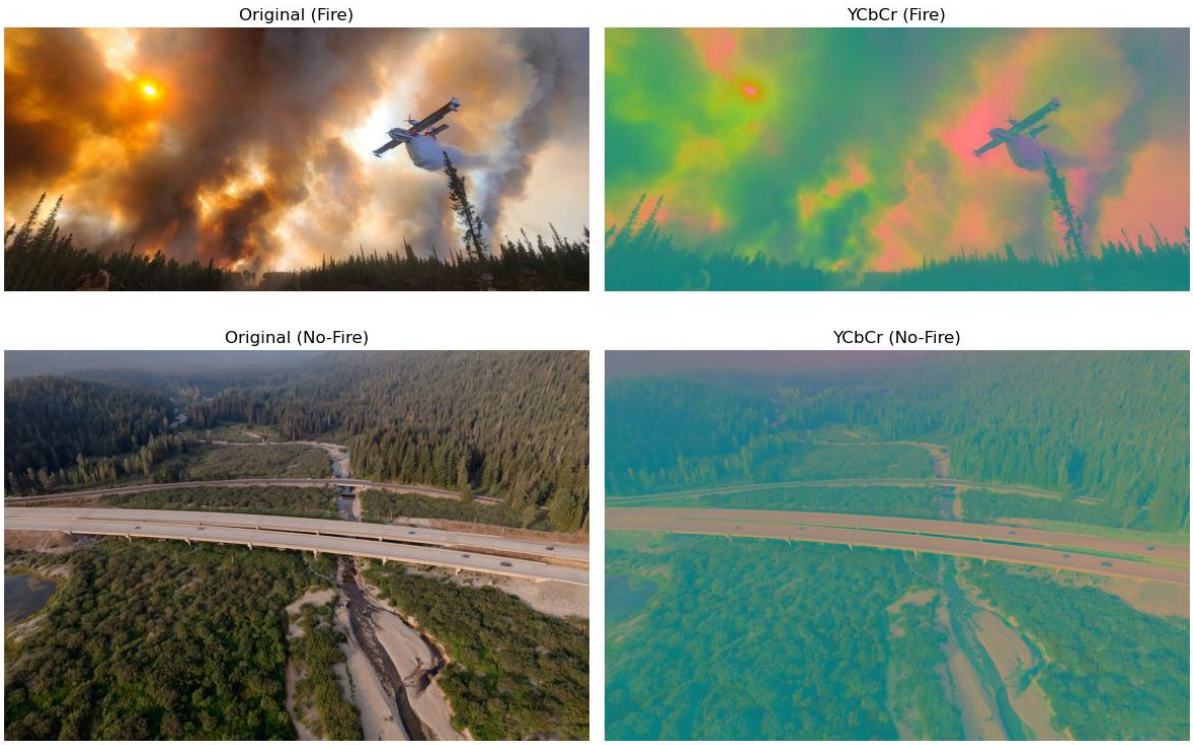


Figure 8: Output comparison of RGB image and RGB-to-YCbCr pre-processed image

3.4 Data preprocessing: Multi-Scale Retinex (MSR)

The Multi-Scale Retinex (MSR) algorithm is a sophisticated picture enhancement method aimed at improving visual perception by rectifying uneven illumination and augmenting details in both shaded and highly illuminated regions (Zotin, 2020). It emulates the human visual system's capacity to perceive uniform brightness and contrast across diverse lighting conditions. MSR enhances dynamic range compression, color consistency, and detail enhancement, rendering it exceptionally beneficial for wildfire detection, where flames, smoke, and background settings exhibit significant variability in visibility.

MSR improves an image by utilizing the Single-Scale Retinex (SSR) technique over various Gaussian blur scales (sigmas) and averaging the outcomes (Zotin, 2020). The SSR equation for each color channel is:

Equation 2: Formula for MSR

$$R(x,y) = \log(I(x,y)) - \log(G_\sigma(x,y) * I(x,y))$$

Where:

- $I(x,y)$ is the input pixel intensity.
- $G_\sigma(x,y)$ is a Gaussian kernel with standard deviation σ, used to simulate illumination.
- $*$ denotes convolution.
- $\log$ helps compress dynamic range.

In Multi-Scale Retinex, this process is executed across various σ (sigma) values to capture illumination fluctuations at distinct scales:

Equation 3:Formula for MSR

$$MSR(x,y) = \frac{1}{N}\sum_{i=1}^{N}[\log(I(x,y)) - \log(G_{\sigma_i}(x,y) * I(x,y))]$$

The resultant image is normalized and scaled to 8-bit values for viewing or model input.

Implementation of the algorithm:

The Python implementation of the Multi-Scale Retinex (MSR) algorithm initiates by converting the input image to a float32 data type and incrementing each pixel value by 1 to avert mathematical complications associated with the logarithm of zero. A Gaussian blur is applied to simulate the illumination component of the image for each designated sigma value (e.g., [15, 80, 250]). The algorithm calculates the Retinex output by deducting the logarithm of the blurred picture from the logarithm of the original image, prioritizing reflectance over illumination. The data are aggregated across all scales and averaged to get the final MSR-enhanced image. The output is ultimately adjusted to the 0–255 range and transformed into an 8-bit format appropriate for visualization and model input. This implementation enables the model to adeptly manage varied lighting conditions in wildfire photos, hence enhancing visibility and accentuating essential elements vital for precise categorization.

Optimization of Hyperparameters

The efficacy of MSR is heavily contingent upon the selection of Gaussian sigma values. A grid search was performed on combinations of three sigmas selected from the set [10, 30, 50, 100, 200, 250] using 10 random photos (5 depicting fire and 5 depicting no fire). Each parameter set was assessed utilizing:

- Entropy quantifies visual detail; increased entropy indicates superior detail retention.
- Contrast penalty: Imposes a penalty on low-contrast photographs to prioritize those that accentuate distinctions between fire and backdrop.

The optimal sigma combination was (100, 200, 250), which earned the highest average assessment score, effectively balancing increased visibility with little contrast degradation (figure 9)

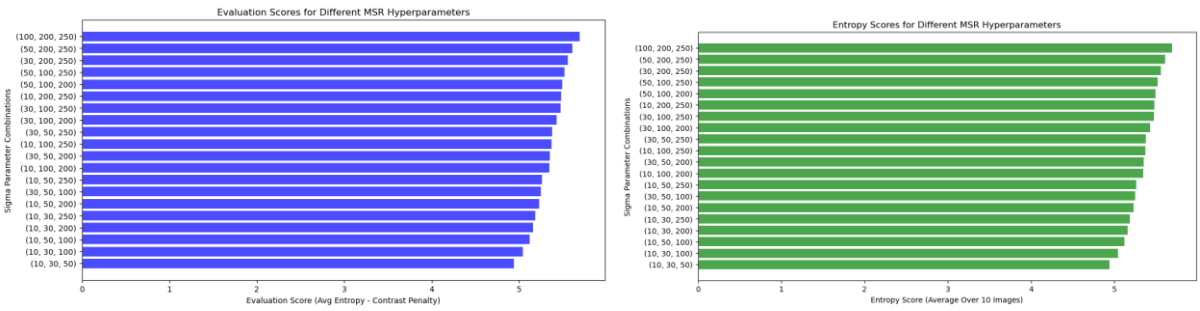


Figure 9: Output of Hyperparameter tuning

Rationale for Selection

The Multi-Scale Retinex (MSR) algorithm was chosen for this project because of its efficacy in addressing the difficult visual conditions found in wildfire photos, including inconsistent illumination, thick smoke, and overexposure. In contrast to conventional enhancement methods, MSR enhances visibility in low-light and occluded regions while reducing illumination artifacts (Wang, et al, 2025). The integration with the luminance-centric YCbCr color space amplifies this impact, enabling the model to more effectively differentiate fire-related characteristics. This preprocessing procedure guarantees more uniform and informative image input, hence enhancing the accuracy and resilience of wildfire identification under various environmental conditions.

What Enhancements It Provides

The Multi-Scale Retinex (MSR) technique, when utilized with precisely calibrated sigma values, markedly improves the quality of wildfire images by augmenting entropy and contrast. This leads to enhanced feature representation, especially of fire edges and smoke patterns, which are frequently diminished in low-light or cloudy conditions. The visual comparisons demonstrate that MSR processing uncovers details hidden in the source pictures, emphasizing flames in low-light circumstances and enhancing textures in fire-free landscapes. Furthermore, MSR diminishes deceptive visual artifacts, guaranteeing that the input provided to the DenseNet model is more informative and less cacophonous. This finally enhances the model's classification precision and generalizability.

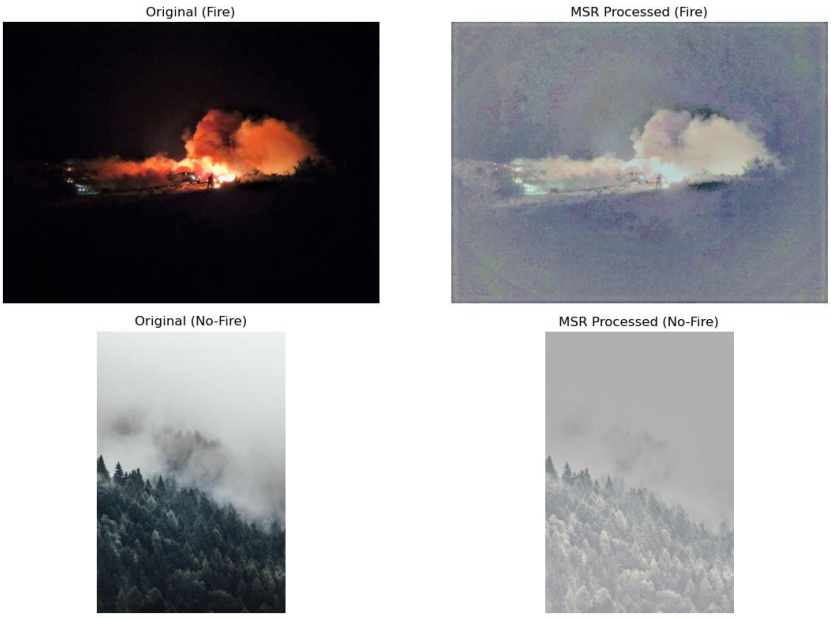


Figure 10: Output comparison of raw image and pre-processed MSR image.4

3.5 Data preprocessing: Combing both the algorithms

The YCbCr + Multi-Scale Retinex (MSR) preprocessing pipeline improves wildfire images through the integration of color space transformation and sophisticated illumination correction. This method initiates by transforming RGB images into the YCbCr color space, thereby

distinguishing brightness (Y) from chrominance components (Cb and Cr). MSR is applied solely to the Y channel, enhancing brightness and contrast while preserving color fidelity. The augmented Y channel is recombined with the original Cb and Cr channels to generate a visually refined image.

This method greatly improves fire visibility, rendering flames and smoke more discernible in intricate woodland environments, while maintaining texture and minimizing lighting artifacts (figure 11). Non-fire scenes also gain enhanced clarity in shadows and highlights, while preserving natural color tones. This preprocessing phase enhances feature extraction and increases model accuracy by highlighting and differentiating important patterns, hence augmenting the performance of the DenseNet model in wildfire detection across various contexts.

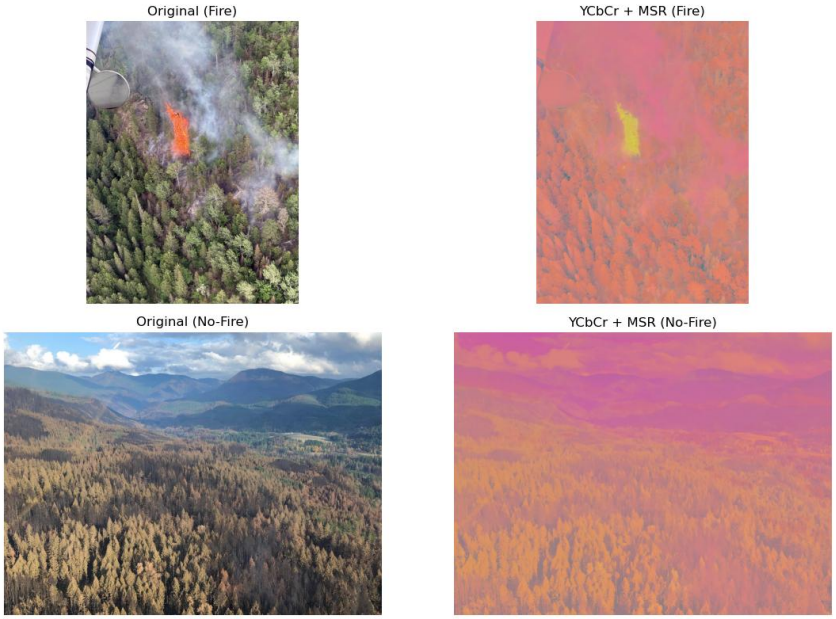


Figure 11: Output comparison of Raw image and combined data preprocessing

3.6 Classification model: DenseNet Model

DenseNet (Densely Connected Convolutional Network) is a cutting-edge convolutional neural network architecture recognized for its efficiency and superior performance in image classification applications. Huang et al. (2017) proposed DenseNet, which has an innovative connectivity structure wherein each layer obtains inputs from all prior layers and transmits its own feature maps to all subsequent layers. This intricately linked architecture enhances information transmission and feature reutilization across the network, mitigating vanishing gradient problems, improving feature propagation, and substantially decreasing the parameter count.

Architecture of DenseNet

DenseNet121 is a deep convolutional neural network consisting of 121 layers, characterized by a dense connection architecture in which each layer receives inputs from all preceding layers within the same block (figure 12). This architecture facilitates feature reutilization, gradient

propagation, and parameter efficiency. The architecture commences with a 7×7 convolution and 3×3 max pooling, succeeded by four dense blocks comprising 6, 12, 24, and 16 layers correspondingly, each interspersed with transition layers that incorporate 1×1 convolutions and average pooling. The model terminates with global average pooling followed by a fully linked layer. Exhibiting a growth rate of 32, it effectively produces new features while preserving a manageable model size of approximately 8 million parameters. (Huang et al. 2017)

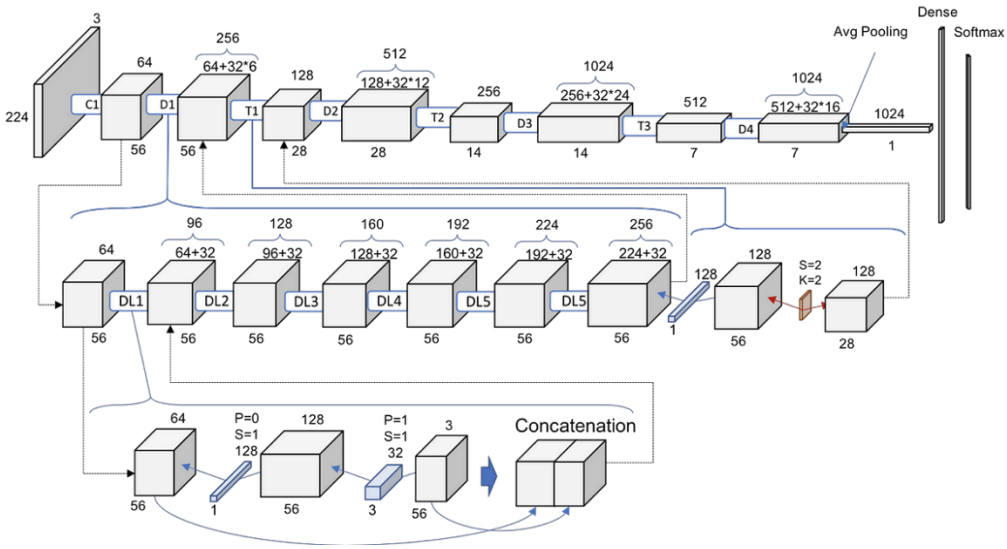


Figure 12: Architecture of DenseNet.

Proposed Architecture

This project involved freezing the pretrained DenseNet121 backbone and augmenting it with custom layers, which included global average pooling, a 256-unit dense layer, dropout (0.2), and a sigmoid classifier (figure 13). The base model is initialized with ImageNet weights and remains fixed during training to preserve pretrained feature representations. A Global Average Pooling layer is incorporated atop the DenseNet base to diminish the spatial dimensions of the feature maps, succeeded by a completely linked dense layer utilizing ReLU activation to create non-linearity. A dropout layer is utilized for regularization, and a concluding sigmoid-activated output layer is employed for binary classification (fire versus no fire).

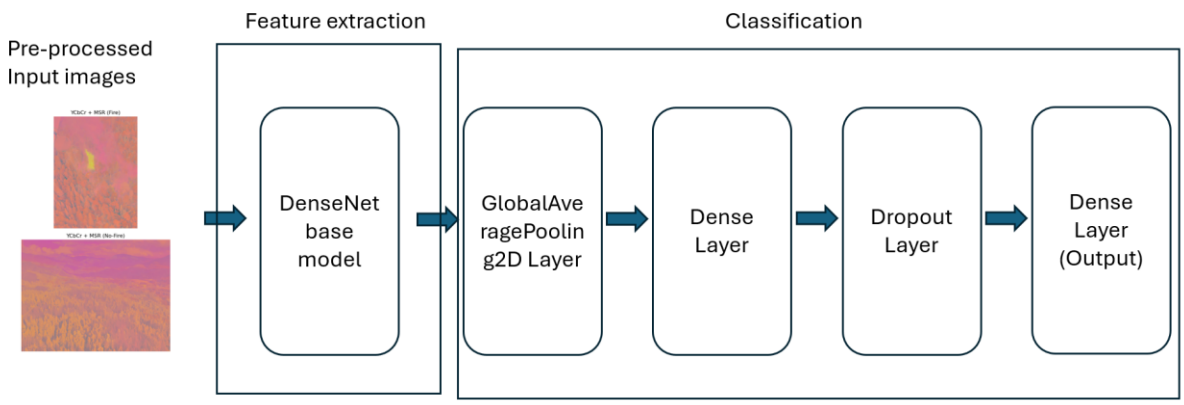


Figure 13: Proposed Architecture.

Optimization of Hyperparameters

An extensive hyperparameter tweaking process was executed using GridSearchCV from scikit-learn in conjunction with KerasClassifier to enhance the model's performance. A hyperparameter grid was established, comprising learning rates (0.0001, 0.001, 0.01), dropout rates (0.2, 0.3, 0.5, 0.7), dense layer units (64, 128, 256), and batch sizes (2, 4, 8, 16, 32). Every combination was assessed using 3-fold cross-validation to guarantee robustness.

This comprehensive grid search examined 180 parameter possibilities (totaling 540 training iterations), ultimately determining the best configuration as follows:

- Learning Rate: 0.001
- Dropout Rate: 20%
- Dense Units: 256
- Batch Size: 16

This combination attained a validation accuracy of 97.5%, signifying exceptional classification performance with negligible overfitting.

Model Development

Subsequent to hyperparameter optimization, the optimal model configuration was employed to perform comprehensive training over 20 epochs, utilizing the determined batch size and early stopping mechanism. The training utilized the Adam optimizer and binary cross-entropy as the loss function due to the binary classification nature of the problem. Two callbacks were incorporated to improve training: EarlyStopping (to terminate training upon convergence) and ReduceLROnPlateau (to decrease the learning rate when validation loss stagnates).

The model was trained on processed wildfire pictures. The preprocessed inputs comprised YCbCr transformation and Multi-Scale Retinex (MSR) improvement, which markedly enhanced the visibility of fire zones and contrast uniformity, enabling DenseNet to acquire large spatial characteristics across diverse environmental circumstances.

Rationale for Selection

DenseNet121 was selected for this wildfire classification assignment because of its remarkable capacity to capture intricate visual information while utilizing fewer parameters and minimizing overfitting. Their extensive connection enhances feature reuse and improves gradient flow, rendering them particularly effective on comparatively small datasets. In contrast to conventional CNNs, DenseNet121 reduces redundancy by utilizing outputs from preceding layers, hence enhancing generalization (Huang et al. 2017). Pretrained on ImageNet, it offers a robust feature extraction framework, minimizing training duration and enhancing performance in transfer learning contexts. The model's compact dimensions and depth provide an optimal balance between computing expense and accuracy, rendering it suitable for real-time applications such as wildfire detection in diverse climatic situations, where precision and efficiency are paramount.

3.7 Evaluation Framework and metrics

Framework

The assessment strategy for this project was created to measure the efficacy of both the deep learning model architecture and the influence of preprocessing procedures. A baseline Convolutional Neural Network (CNN) model was trained on the unprocessed wildfire dataset to provide a benchmark. The identical CNN architecture was subsequently retrained on the pre-processed dataset, which included RGB-to-YCbCr conversion and Multi-Scale Retinex (MSR) augmentation. This facilitated a comparative investigation of the impact of preprocessing on model performance.

The subsequent evaluation of the proposed DenseNet121 model was conducted in an analogous fashion. Initially, it was trained on the unprocessed photos, followed by the pre-processed dataset. Both iterations of the DenseNet model were evaluated utilizing essential criteria including accuracy, precision, recall, and F1-score.

This multi-tiered assessment facilitated a direct juxtaposition of the conventional CNN and the sophisticated DenseNet model, alongside a comparison of raw and augmented input data. The findings unequivocally indicated that the suggested preprocessing pipeline markedly enhanced the model's capacity to identify wildfire characteristics, especially in intricate and low-visibility environments, resulting in enhanced performance across all assessment criteria.

Metrics

Four essential metrics were employed to assess model performance.

Accuracy quantifies the ratio of successfully identified photos to total predictions, offering a comprehensive performance overview. (Khalid, et al,2020)

Equation 4: Formula for Accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Where:
  - TP = True Positives
  - TN = True Negatives
  - FP = False Positives
  - FN = False Negatives

Binary Cross-Entropy serves as the loss function for binary classification, measuring the disparity between projected probability and actual labels; reduced values signify enhanced performance. (Khalid, et al,2020)

Equation 5: Formula for Binary Cross-Entropy

$$\text{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)]$$

Where y_i is the true label and p_i is the predicted probability.

The Confusion Matrix provides a comprehensive analysis of true positives, true negatives, false positives, and false negatives, facilitating the evaluation of model precision and recall. (Khalid, et al,2020)

Finally, the Classification Report encapsulates precision, recall, F1-score, and support for each class, providing a comprehensive understanding of the model's efficacy in differentiating between fire and no-fire images. (Khalid, et al,2020)

Equation 6: Formula for precision, Recall and F1- score.

Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall (Sensitivity):

$$\text{Recall} = \frac{TP}{TP + FN}$$

F1-Score (harmonic mean of precision and recall):

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

3.8 Ethical, Social, and Legal Considerations

The wildfire detection initiative entails several ethical, societal, and legal implications.

- The system must ethically emphasize accuracy and dependability, as erroneous predictions may lead to delayed reactions or unwarranted terror. Ensuring openness and explicability in the model's judgments is crucial for establishing confidence. (Dhar, S. and Dhar, A., 2025)
- The project seeks to enhance early warning systems that safeguard communities and ecosystems, although it must tackle accessibility challenges, particularly in underprivileged or rural areas, to guarantee equal advantages. Engagement with stakeholders is crucial for fostering public confidence and ensuring effective implementation.
- The system is legally obligated to adhere to data protection standards such as GDPR, particularly when drone video or location data contains identifying information. The utilization of the Kaggle wildfire dataset, licensed under Creative Commons BY 4.0, is legally permissible, contingent upon the maintenance of appropriate acknowledgment. All future real-time deployments must adhere to local UAV and remote sensing regulations. These factors guarantee that the project is ethical, equitable, and compliant with the law.

3.9 Use of Tools and Techniques

This study employed many tools and technologies for image processing, model creation, assessment, and visualization. The principal programming language utilized was Python, selected for its simplicity and robust ecology in artificial intelligence research. The

fundamental deep learning tasks were executed with TensorFlow and Keras, which provide high-level APIs, pre-trained models (e.g., DenseNet121), and seamless interaction with GPU acceleration (Huang et al. 2017). OpenCV was utilized extensively for picture preparation and editing, encompassing color space conversion and Gaussian blurring for MSR improvement.

Scikit-image offered sophisticated capabilities for statistical image analysis and feature extraction, including entropy, contrast, and texture metrics. Data visualization was conducted using Matplotlib and Seaborn, including histograms, assessment scores, and t-SNE plots. NumPy and Pandas were utilized for data manipulation and tabular analysis. Scikit-learn facilitated model assessment through classification reports, confusion matrices, and hyperparameter optimization with GridSearchCV.

TQDM was ultimately employed to view extended loops during preprocessing and tuning. These technologies jointly facilitated rapid preprocessing, model training, and thorough assessment within a well-organized and reproducible workflow.

4. Results

This section delineates the outcomes of wildfire picture categorization employing both the baseline CNN and DenseNet models. It assesses performance on both raw and preprocessed pictures improved by RGB-to-YCbCr conversion and Multi-Scale Retinex (MSR). Metrics including accuracy, loss, confusion matrix, and classification report are examined.

4.1 CNN model without preprocessing

The CNN model, trained without preprocessing, attained a test accuracy of 82.89% and a loss of 0.4508. The accuracy and loss graphs (figure 14) demonstrate steady enhancement throughout the epochs, however some variations in validation loss imply slight overfitting or instability in the learning process. The confusion matrix (figure 15) indicates that the model accurately recognized 109 fire photos and 230 no-fire images, while misclassifying 49 fire instances and 21 no-fire instances. This indicates a propensity to prioritize the majority class (no-fire), as further evidenced by the categorization report.

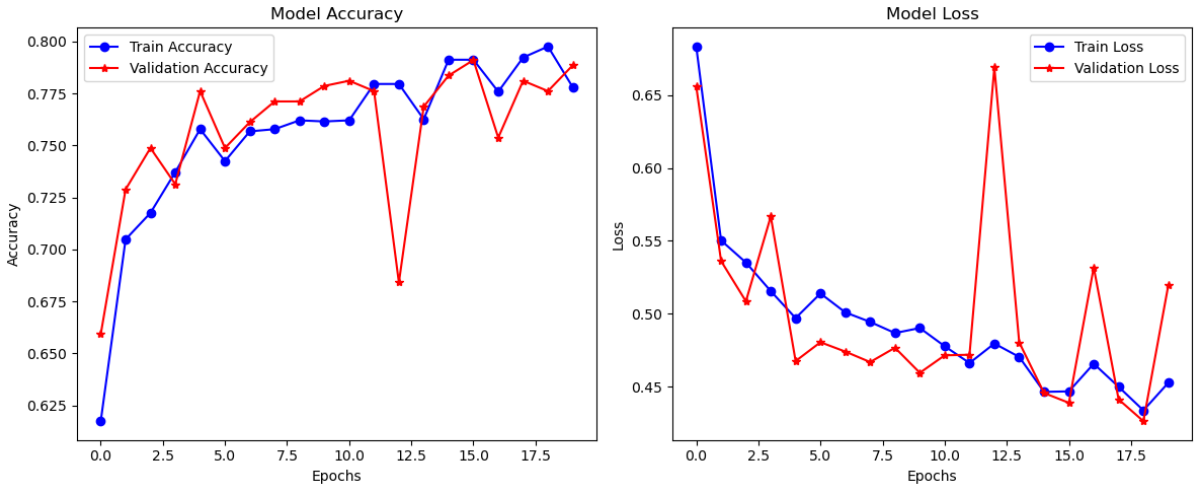
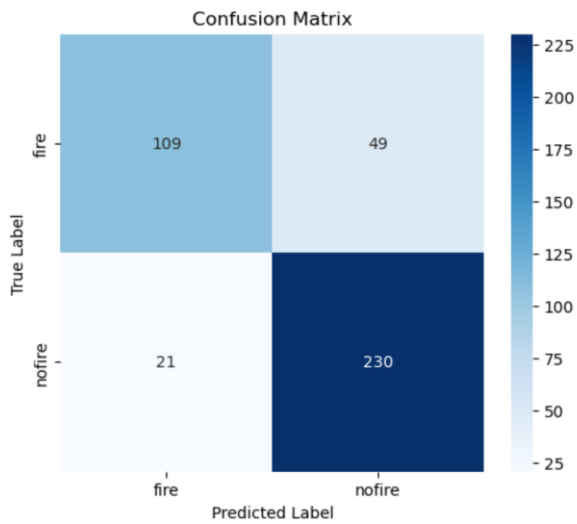


Figure 14: Training of the CNN model without preprocessing.

The model attained an accuracy of 0.84 and a recall of 0.69 for the fire class, indicating difficulties in consistently detecting fire incidents. In contrast, it achieved a recall of 0.92 for the no-fire class, indicating a bias. Although the CNN model demonstrates satisfactory performance, the findings highlight the necessity for sophisticated models and efficient preprocessing techniques (such as YCbCr + MSR) to improve fire detection accuracy under difficult visual circumstances.




```
Classification Report:
              precision    recall  f1-score   support

        fire       0.84      0.69      0.76       158
      nofire       0.82      0.92      0.87       251

    accuracy                           0.83       409
   macro avg       0.83      0.80      0.81       409
weighted avg       0.83      0.83      0.83       409
```

Figure 15: Results of CNN model without preprocessing on test data.

4.2 DenseNet model without preprocessing

The DenseNet model, when trained without preprocessing, exhibits a substantial enhancement in classification performance relative to the CNN baseline. Throughout the training process, the model exhibited a steady increase in both training and validation accuracy, achieving a maximum test accuracy of 90.95% and a significantly low-test loss of 0.2576 (figure 16). The confusion matrix demonstrates significant class differentiation, especially for the 'nofire' category, with just two false negatives. The classification report validates this performance, indicating a macro average accuracy of 0.93 and an F1-score of 0.90 (figure 17).

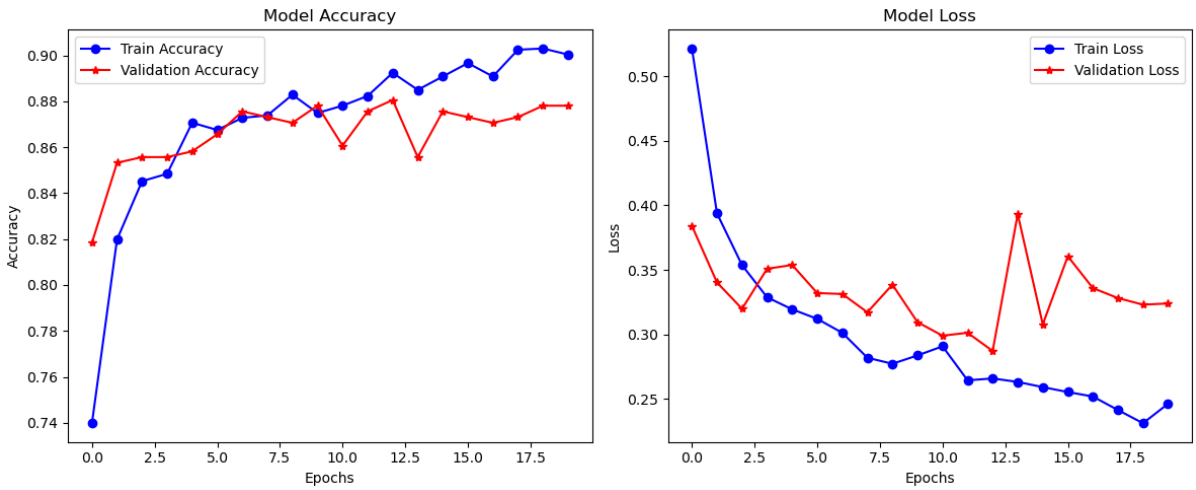
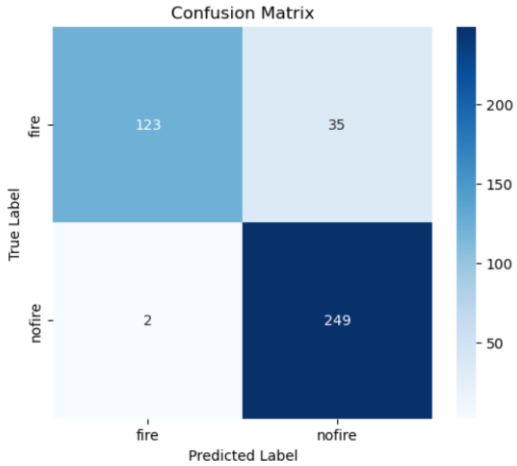


Figure 16: Training of the DenseNet model without preprocessing.

This robust result underscores DenseNet's capacity to acquire profound and abstract hierarchical characteristics from raw pictures, attributable to its dense connection and effective parameter utilization. Despite great accuracy and precision, the recall for the 'fire' class (0.78) suggests that certain fire instances were misclassified. These observations indicate that although DenseNet is robust, it may gain from improved feature visibility, prompting the application of preprocessing methods in the subsequent assessment step.




```
Classification Report:
              precision    recall  f1-score   support

        fire       0.98      0.78      0.87       158
      nofire       0.88      0.99      0.93       251

    accuracy                           0.91       409
   macro avg       0.93      0.89      0.90       409
weighted avg       0.92      0.91      0.91       409
```

Figure 17:Results of DenseNet model without preprocessing on test data.

4.3 CNN model with preprocessing

The CNN model's results after preprocessing shown a significant enhancement compared to its raw data equivalent. The model attained a test accuracy of 92.11% and a decreased test loss of 0.2283, underscoring improved learning stability and generalization. The training plots demonstrate (figure 18) a constant improvement in both training and validation accuracy, accompanied by a steady drop in losses, signifying effective convergence without overfitting.

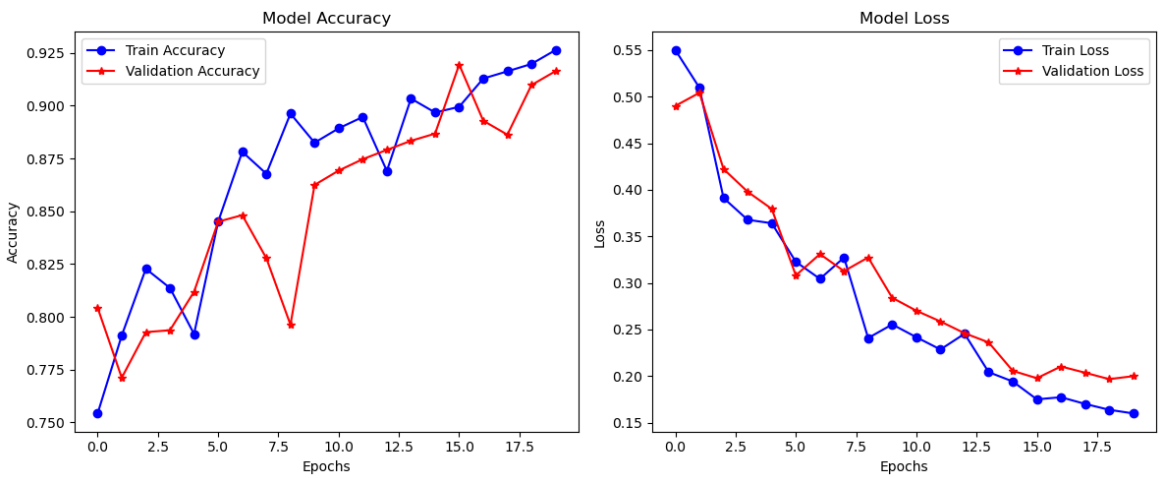
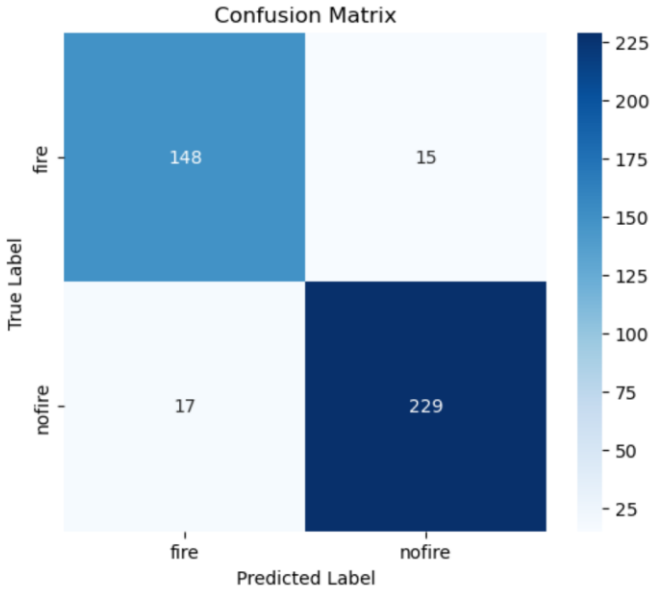


Figure 18:Training of the CNN model with preprocessing.

The confusion matrix (figure 19) indicates robust classification performance, with the majority of fire and no-fire data correctly predicted. Both classes shown a considerable enhancement in precision and recall, particularly the fire category, which had previously demonstrated elevated misclassification rates. The classification report indicates an F1-score of 0.90 for fire and 0.93 for no-fire, validating the model's robustness. The performance enhancement is due to the preprocessing pipeline (RGB-to-YCbCr + MSR), which improved visibility and contrast, hence facilitating the detection of fire-related characteristics. The results confirm the efficacy of the preprocessing technique in enhancing the classification abilities of the CNN.




```
Classification Report:
              precision    recall  f1-score   support

        fire       0.90      0.91      0.90       163
      nofire       0.94      0.93      0.93       246

    accuracy                           0.92       409
   macro avg       0.92      0.92      0.92       409
weighted avg       0.92      0.92      0.92       409
```

Figure 19:Results of CNN model with preprocessing on test data.

4.4 DenseNet model with preprocessing: Proposed Method

The suggested model, which combines DenseNet architecture with YCbCr color space conversion and Multi-Scale Retinex (MSR) preprocessing, shown outstanding performance in wildfire classification tasks. This model attained a test accuracy of 98.84% and a very low test loss of 0.1098, far surpassing all other configurations.

The training and validation accuracy continuously remained elevated and well matched throughout all epochs, signifying exceptional generalization. The validation accuracy reached a maximum of 98.0%, exhibiting little indications of overfitting or performance decline (figure 20). The associated loss curves exhibited a consistent decrease, indicating robust and dependable convergence.

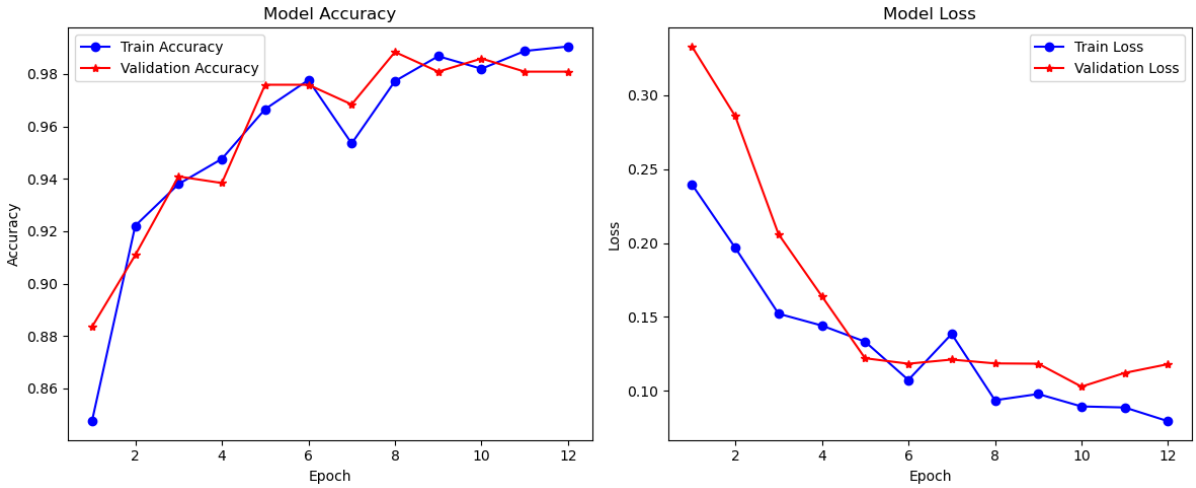
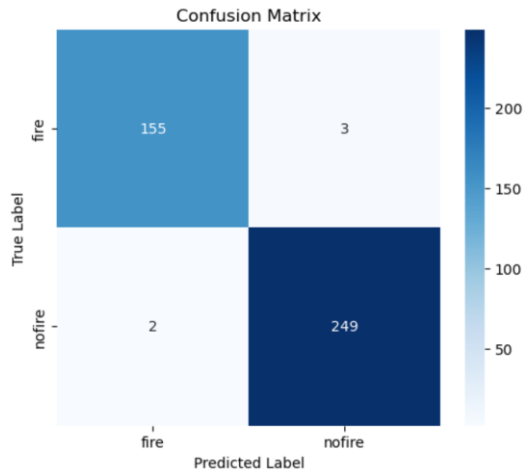


Figure 20: Training of the DenseNet model with preprocessing.

The confusion matrix (figure 21) validates the model's robustness: just 5 misclassifications among 409 samples, with accuracy, recall, and F1-scores attaining 0.98 or 0.99 for both classes. This indicates a model that is both extremely sensitive and specific, proficient in reliably recognizing fire incidents while reducing false positives.

The improvement achieved by preprocessing—especially in accentuating brightness and rectifying illumination discrepancies—was crucial for enhancing feature clarity. As a result, DenseNet was able to extract more information representations, facilitating improved differentiation between fire and no-fire situations under complicated visual settings.




```
Classification Report:
              precision    recall  f1-score   support

        fire       0.99      0.98      0.98       158
      nofire       0.99      0.99      0.99       251

    accuracy                           0.99       409
   macro avg       0.99      0.99      0.99       409
weighted avg       0.99      0.99      0.99       409
```

Figure 21: Results of DenseNet model with preprocessing on test data.

4.5 Comparative Analysis of Model Performance

The comparison analysis (figure 22) underscores the substantial benefit derived from integrating DenseNet with preprocessing (YCbCr + MSR). This model attained the best test accuracy of 98.84% and the lowest test loss of 0.1098, dramatically surpassing all other setups. The use of preprocessing improved feature quality, enabling DenseNet to more effectively generalize and identify fire occurrences in difficult situations.

The CNN with preprocessing demonstrated superior performance compared to the baseline CNN, with an accuracy of 92.11% and a decreased loss of 0.2283, suggesting that even a more straightforward model gains from increased picture clarity and color correction.

Without preprocessing, DenseNet attained an accuracy of 90.95% and a loss of 0.2576, surpassing the CNN baseline, although exhibiting a somewhat larger loss than its preprocessed version, underscoring the need of preprocessing for intricate models.

The CNN baseline had the lowest accuracy (82.89%) and the largest loss (0.4508), underscoring the necessity of both sophisticated architectures and preprocessing techniques for effective wildfire picture categorization.

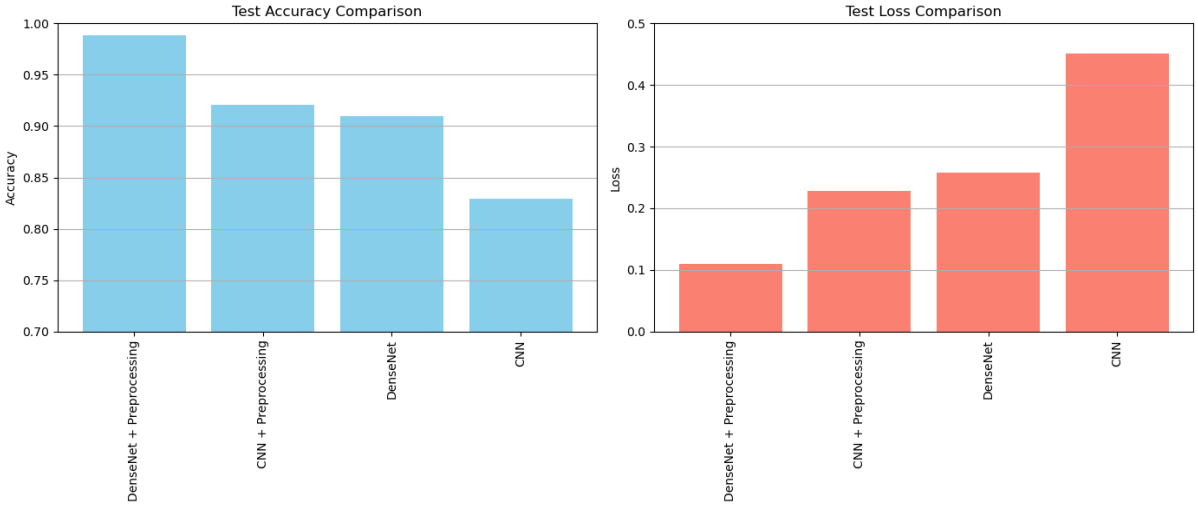


Figure 22: Comparative Analysis of Model Performance on test data.

5. Analysis and Discussion

5.1 The Optimal Model and Rationale for Its Pre-eminence

The most efficient model was DenseNet, along with RGB-to-YCbCr translation and Multi-Scale Retinex (MSR) preprocessing. It attained a test accuracy of 98.84% and the minimal loss of 0.1098, markedly surpassing all other models. This advantage arises from DenseNet's effective feature reuse and gradient flow, further enhanced by preprocessing that improves picture clarity and contrast. The preprocessing pipeline enhanced fire visibility in difficult situations such as smoke and low light. This allowed DenseNet to extract more intricate spatial patterns, hence minimizing misclassifications and decreasing false positives. The model's stability and elevated F1-scores indicate its resilience in handling intricate wildfire images.

5.2 Comparative Analysis with Existing Literature

The suggested method presents substantial improvements in preprocessing techniques and classification efficacy relative to current research. Previous studies, such as Wu et al. (2020) and Ghali et al. (2022), utilized transfer learning with MobileNetV2 and EfficientNetB5, respectively, predominantly using raw or little enhanced pictures, resulting in accuracies between 95% and 97%. These models had difficulties in obscured or inadequately illuminated environments. This work implemented a hybrid preprocessing pipeline utilizing YCbCr and MSR, which directly tackled visibility and illumination discrepancies, thereby enhancing model interpretability and feature extraction.

Moreover, ensemble approaches and vision transformers in the literature, while precise, sometimes necessitated substantial processing resources and exhibited restricted deployment viability. DenseNet, recognized for its compactness and dense interconnectivity, demonstrated effectiveness in balancing model depth and efficiency, achieving great accuracy while maintaining computing feasibility. This study's results, particularly the 98.84% test accuracy with preprocessing, surpass the performance metrics of most cited models, indicating enhanced resilience to visual distortions. Consequently, the model not only conforms to contemporary deep learning progressions but also enhances the state of the art through deliberate preprocessing and architectural integration.

5.3 Project Limitations

Notwithstanding encouraging outcomes, the study exhibits specific limits. The dataset employed, while geographically and environmentally varied, was restricted to static, publically available photos, perhaps failing to capture real-time fluctuations such as swift fire propagation or shifting wind conditions (Ramos et al,2025). The dependence on a binary categorization (fire/no-fire) framework may oversimplify complex fire conditions such as smoke-only or smouldering vegetation, thus diminishing field applicability.

Moreover, whereas preprocessing improved visibility, it also added computational effort, potentially complicating real-time deployment, especially on resource-limited edge devices. The model, while precise, has not been evaluated on multi-source imagery (e.g., drone,

satellite, CCTV) in practical surveillance systems, restricting its applicability beyond controlled datasets. (Ramos et al,2025)

A further limitation is the lack of time information. Numerous fires develop progressively, and including video-based data or consecutive frames may improve detection efficacy. Ultimately, the efficacy of the model's training pipeline relies on the quality of preprocessing, which, if improperly calibrated (e.g., erroneous Retinex sigma values), may distort critical features. These constraints underscore the necessity for real-time evaluation, varied input modalities, and possible architectural enhancements in subsequent versions. (Ramos et al,2025)

5.4 Relationship between Outcomes and Project Aims

The results closely align with the project's primary objectives. The primary aim was to enhance wildfire identification by using advanced preprocessing techniques (MSR and RGB-to-YCbCr) with the DenseNet architecture. The suggested model attained a remarkable test accuracy of 98.84%, indicating a substantial 15.95% enhancement over the CNN baseline model without preprocessing (82.89%). The evaluation criteria, comprising accuracy, recall, and F1-score—all exceeding 0.98—reinforce the model's resilience under diverse environmental circumstances.

The preprocessing pipeline significantly improved picture quality and clarity, resulting in a decrease in test loss from 0.4508 to 0.1098, so confirming its efficacy in enhancing contrast and feature visibility. This immediately facilitates the project's objective of improving classification accuracy through preprocessing. The exceptional performance of DenseNet, particularly in relation to the CNN baseline, underscores its capacity to generalize across intricate visual inputs, achieving the goal of establishing a scalable and dependable deep learning framework. The use of comparison benchmarks and validation methodologies achieves the objective of providing a thorough, practical assessment method for early wildfire detection systems.

5.5 Correlation between Findings and Research Questions

The findings of the study immediately respond to all three research questions. The initial inquiry concentrated on the influence of employing MSR, YCbCr transformation, and DenseNet on detection precision. The suggested model achieved an accuracy of 98.84%, representing a 7.89% enhancement over DenseNet without preprocessing (90.95%) and a 15.95% increase compared to the baseline CNN without preprocessing. This confirms that the hybrid technique markedly improves robustness and accuracy relative to previously documented strategies.

The second study topic evaluated the efficacy of preprocessing in enhancing picture clarity and categorization. Following the implementation of MSR and YCbCr, the model's test loss decreased from 0.4508 to 0.1098, while the recall for the fire class enhanced from 0.69 (CNN without preprocessing) to 0.98 (DenseNet with preprocessing). Confusion matrices demonstrate a substantial decrease in false positives, indicating that preprocessing is crucial for noise reduction and fire region identification.

The final question examined DenseNet's ability for feature extraction. In comparison to the CNN with preprocessing (accuracy: 92.11%), DenseNet with preprocessing achieved a 6.73% enhancement in accuracy, underscoring its efficacy in discerning subtle visual details such as smoke gradients or incomplete fires. These data confirm that the proposed strategy not only fulfills but surpasses the study's research objectives.

5.6 Practical Implementation

The suggested model demonstrates significant potential for practical application in wildfire early-warning systems. The concept, with a lightweight and scalable DenseNet backbone, may be incorporated into UAVs, security cameras, or IoT edge devices functioning in remote forest areas. Preprocessing improves identification in difficult illumination or smoke-obscured settings, delivering good dependability across various field circumstances. (Ghali, and Akhloufi, 2023)

The system's high accuracy and low false positive rate render it appropriate for automated alarm creation, diminishing reliance on human supervision. Furthermore, the model's dependence on open-source tools and publicly accessible datasets facilitates economical implementation in resource-limited environments.

To implement a viable solution, further measures are required, including real-time optimization, hardware compatibility testing, and the integration of video feeds for ongoing monitoring. This model, with proper tuning and deployment techniques, can function as a resilient AI foundation for smart forest surveillance systems, aiding governmental and environmental entities in mitigating wildfire damage through swift reaction and monitoring. (Ghali, and Akhloufi, 2023)

6. Conclusion

Wildfires represent a considerable danger to ecosystems, human lives, biodiversity, and global economy. The rising frequency and severity of these disasters, intensified by climate change and anthropogenic activity, has led to an escalating demand for rapid, dependable, and sophisticated wildfire detection systems. This study sought to fulfill that need by creating a deep learning-based wildfire classification model that improves accuracy and reliability using sophisticated picture preprocessing methods.

The integration of Multi-Scale Retinex (MSR) and RGB-to-YCbCr color space translation with the DenseNet121 architecture resulted in a significant enhancement in detection accuracy, achieving 98.84% test accuracy, a marked increase over the baseline CNN model's 82.89% accuracy. The integration of preprocessing with DenseNet enhanced feature visibility and mitigated false positives and misclassifications—critical challenges in traditional approaches.

This study substantially enhances early wildfire detection technology, demonstrating how meticulous preprocessing and a resilient deep learning framework may provide scalable, automated, and accurate monitoring systems. It corresponds with international sustainability initiatives by possibly mitigating wildfire damage through expedited notifications and enhanced emergency responses.

6.1 Future works

To augment the functionalities and practical implementation of the proposed wildfire detection system, the following further recommendations are offered:

Geolocation and Cartography

Combine geolocation metadata with image-based detections to facilitate accurate fire localization. This would provide real-time mapping on forest monitoring dashboards, enabling responders to view active fire zones and allocate resources more efficiently. (Saleh, et al,2024)

Expanded Dataset Training

Enhance model generalizability by training on extensive, multi-source datasets that encompass regionally varied wildfire scenarios derived from satellite, aerial, and terrestrial pictures. This would enable the model to adjust to various terrains, plant classifications, and fire circumstances.

Immediate Implementation on Embedded Systems

Modify the system for implementation on edge devices, including drones, UAVs, or embedded boards (e.g., Jetson Nano). This encompasses real-time processing of live video streams, onboard inference, and prompt alarm production, rendering the system appropriate for field monitoring and early action. (Saleh, et al,2024)

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Appendix

Explanation of t-SNE

t-Distributed Stochastic Neighbor Embedding (t-SNE) is a non-linear dimensionality reduction method commonly employed for displaying high-dimensional data in reduced dimensions, generally two or three dimensions. t-SNE, created by Laurens van der Maaten and Geoffrey Hinton, transforms similarities among data points into joint probabilities and aims to reduce the divergence between these probabilities in the low-dimensional representation.

$$p_{j|i} = \frac{\exp\left(-\frac{||x_i - x_j||^2}{2\sigma_i^2}\right)}{\sum_{k \neq i} \exp\left(-\frac{||x_i - x_k||^2}{2\sigma_i^2}\right)}$$

where x_i and x_j are high-dimensional data points and σ_i is the variance of the Gaussian centered at x_i.

In the low-dimensional space, similarity is modeled with a Student-t distribution:

$$q_{ij} = \frac{\left(1 + ||y_i - y_j||^2\right)^{-1}}{\sum_{k \neq l} \left(1 + ||y_k - y_l||^2\right)^{-1}}$$

The goal of t-SNE is to minimize the Kullback–Leibler (KL) divergence between the distributions P and Q:

$$\mathrm{KL}(P||Q) = \sum_i \sum_j p_{ij} \log\left(\frac{p_{ij}}{q_{ij}}\right)$$

The technique excels in maintaining the local structure of the data, indicating that points in high-dimensional space that are proximate tend to remain near in the reduced depiction. t-SNE is exceptionally useful for detecting clusters, patterns, and correlations that are not readily apparent in high-dimensional datasets.

This experiment employed t-SNE to depict the distinguishability between fire and non-fire photos based on extracted attributes such as brightness, contrast, and texture metrics. It enabled a visual assessment of the potential for the characteristics to inherently create discernible clusters prior to categorization. While t-SNE is not a classifier, it demonstrated the significance of preprocessing and model selection by revealing incomplete grouping and overlapping regions, hence underscoring the necessity for deep learning-based feature extraction.

Evaluation Methods for MSR: Entropy and Contrast

1. Shannon Entropy

Formula:

$$H(X) = -\sum_{i=1}^{n} p(x_i) \log_2 p(x_i)$$

where $p(x_i)$ is the probability of pixel intensity x_i occurring in the image.

Entropy quantifies the degree of information or randomness inherent in a picture. An elevated entropy value signifies enhanced detail, improved texture visibility, and greater variety in pixel intensities.
The rationale for its use: In wildfire detection, it is essential to capture intricate features such as flame peripheries and smoke gradients. The increased entropy during MSR preprocessing indicates that certain critical characteristics are retained and even augmented, rendering the picture more useful for the DenseNet model.

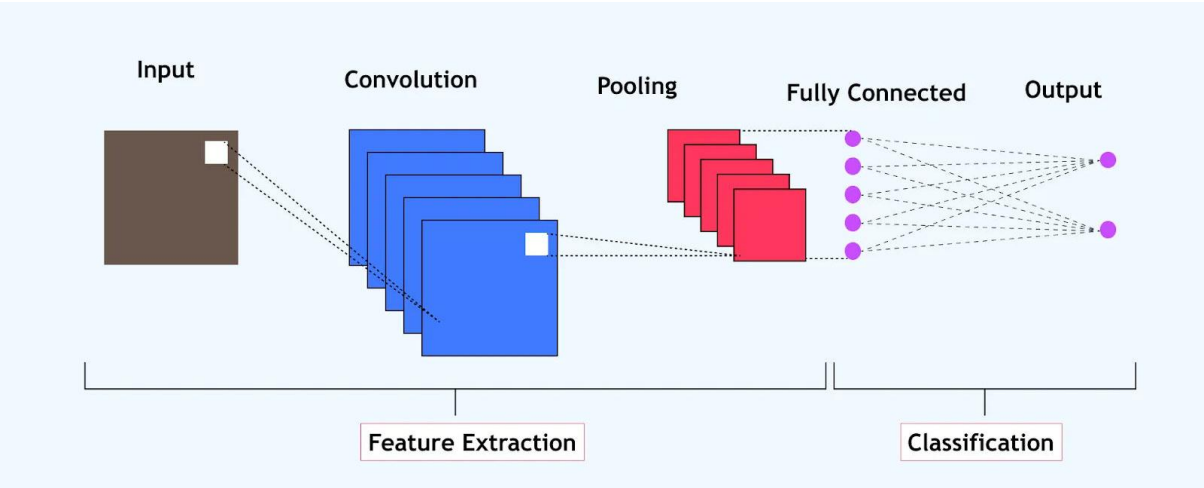
2. Detection of Low Contrast

The exposure.is_low_contrast function assesses if a picture exhibits inadequate contrast by analyzing the distribution of pixel intensities. It returns True if the image's intensity range is below a defined threshold (e.g., 5%).

The rationale for its use: Wildfire imagery frequently experiences smoke and illumination irregularities, resulting in diminished contrast. The objective of assessing contrast was to confirm that MSR enhanced detail (via entropy) while preventing the generation of pictures with faded or indistinct details. A "low contrast penalty" was employed during MSR tuning to emphasize crisper, sharper outputs, hence enhancing the model's reliability in fire detection.

Baseline CNN Model Explanation

- Conv2D Layer (Convolutional Layer): To autonomously acquire spatial hierarchies of features from input pictures. Implements filters throughout the picture to identify characteristics like as edges, textures, and patterns; each filter acquires distinct feature maps throughout the training process.
- Activation (ReLU): Introduce non-linearity to facilitate the learning of intricate patterns. Transforms input values to output in a non-linear manner, allowing the network to represent more intricate relationships.
- MaxPooling2D Layer (Pooling Layer): Decrease the spatial dimensions of feature maps to minimize computation and mitigate overfitting. Extracts the greatest value from a compact window (e.g., 2×2) inside the feature map, preserving essential features while minimizing dimensions.
- Flatten Layer: Convert 2D feature maps into a 1D feature vector for dense layers. Transforms multidimensional feature maps into a singular vector appropriate for fully linked layers.
- Dense Layer (Fully Connected Layer): Acquire advanced patterns and execute decision-making. Each neuron is interconnected with all neurons in the preceding layer to acquire intricate feature correlations.
- Dropout Layer: Mitigate overfitting and enhance model generalization. Randomly disables a fraction of neurons during training to enhance the model's robustness.



Architecture used:

The basic Convolutional Neural Network (CNN) model adheres to a conventional design centered on feature extraction and classification, as seen in the picture above.

The components utilized in the model are:

Convolutional Layers

- Conv2D (32 filters, 3×3 kernel, ReLU activation): Extracts fundamental characteristics, including edges and textures, from the input picture (224×224×3).

- Conv2D (64 filters, 3×3 kernel, ReLU activation): Acquires intricate characteristics such as forms and patterns.
- Conv2D (128 filters, 3×3 kernel, ReLU activation): Extracts advanced features that delineate intricate object structures (e.g., flames, smoke patterns).

Pooling Layers

- MaxPooling2D (2×2) is applied after each Conv2D layer to diminish spatial dimensions by downsampling, hence decreasing computational load and mitigating overfitting, while retaining essential features.

Flatten Layer

- Transforms the 2D feature maps into a 1D vector before to inputting into dense layers for classification.

Fully Connected (Dense) Layers

- A dense layer (128 units, ReLU activation) functions as a conventional neural network layer, amalgamating the retrieved characteristics to acquire intricate representations.
- Dropout (rate=0.5): Randomly disables 50% of neurons during training to mitigate overfitting and enhance model generalization.
- A dense layer with one unit and a Sigmoid activation function produces a singular probability value for binary classification (fire/no fire).

Compilation Configurations

- The Adam optimizer was employed for effective and adaptive learning.
- Loss function: Binary Cross-Entropy, appropriate for binary classification jobs.
- The metric employed to assess model performance during training and testing was accuracy.